

The Constants of Nature

$$\begin{array}{l}
v_{cs} = \frac{E_1}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
eV : Hz = \frac{eV}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
J : Hz = \frac{joule}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
Hz : eV = 2\pi \frac{joule}{eV} \left(\frac{t_p m_p}{s l_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
h = \frac{joule}{eV} \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
h = \left(\frac{l_p m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
h = 2\pi \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
q_e = \pi \left(\frac{l_p^2 m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
2q_c = 2\pi \left(\frac{l_p^2 m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_\theta}{e} = \frac{1}{2} \left(\frac{t_p^2 C m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_N}{e} = \frac{1}{2} \left(\frac{l_p^2 C m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \right) \boxtimes \\
G_0^{-1} = \frac{\pi}{\kappa_1^2} \left(\frac{l_p^2 m_p}{t_p q_p} \right) \left(1 + \sqrt{\zeta(3)} \mathcal{W}_0^2 \right) \boxtimes \\
G_0 = \frac{\kappa_1^2}{\pi} \left(\frac{t_p q_p^2}{l_p^2 m_p} \right) \left(1 - \sqrt{\zeta(3)} \mathcal{W}_0^2 \right) \boxtimes \\
eV = \kappa_1 \left(\frac{q_p}{C} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \right) \boxtimes \\
J : eV = \frac{joule}{q_p} \left(\frac{C}{q_p} \right) \left(1 - \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \right) \boxtimes \\
-\frac{e}{m_e} = -\kappa_1 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \right) \boxtimes \\
\frac{e}{m_e} = \kappa_1 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_\theta}{h} = \frac{1}{4\pi} \text{tesla}^2 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_N}{h} = \frac{\kappa_1}{4\pi} \frac{1}{\text{MHz}} \left(\frac{q_p}{s m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \right) \boxtimes \\
a = \frac{4\pi}{32 s \kappa_1^4} \left(\frac{l_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{W}_0^2 \right) \boxtimes \\
d_{220} = \frac{1}{\sqrt{8}} \frac{4\pi}{32 s \kappa_1^4} \left(\frac{l_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{W}_0^2 \right) \boxtimes \\
R_m = \frac{\kappa_1^4}{4\pi} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \frac{6}{V_0} \right)^2 \mathcal{W}_0^3 \boxtimes \\
E_h : \frac{1}{m} = \frac{2\pi}{\kappa_1^4} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \frac{6}{V_0} \right)^2 \mathcal{W}_0^2 \boxtimes \\
\frac{1}{m} : E_h = \frac{2\pi}{\kappa_1^4} \left(\frac{l_p m_p}{m m_e} \right) \left(1 + \frac{6}{V_0} \right)^2 \mathcal{W}_0^2 \boxtimes \\
A_{90} = \text{ampere} \left(1 + \frac{1}{\delta} \left(\frac{4\pi}{35} \right) \mathcal{W}_0^3 \right) \boxtimes \\
C_{90} = \text{coulomb} \left(1 + \frac{1}{\delta} \left(\frac{4\pi}{35} \right) \mathcal{W}_0^3 \right) \boxtimes \\
\mu_B = \frac{\kappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_p} \right) \left(1 + \delta \left(\frac{K}{6} \right) \mathcal{W}_0^3 \right) \boxtimes \\
\mu_N = \frac{\kappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_p} \right) \left(1 + \delta \left(\frac{K}{6} \right) \mathcal{W}_0^3 \right) \boxtimes \\
\kappa = \left(\frac{l_p^2 m_p}{t_p^2 q_p^2} \right) \left(1 + \left(\frac{C_d}{6} \right) \mathcal{W}_0^4 \right) \boxtimes \\
A_{perm} = \left(\frac{t_p^2 q_p^2}{l_p^2 m_p} \right) \left(1 - \left(\frac{C_d}{6} \right) \mathcal{W}_0^4 \right) \boxtimes \\
\epsilon_0 = \frac{1}{4\pi} \left(\frac{t_p^2 q_p^2}{l_p^3 m_p} \right) \left(1 - \left(\frac{C_d}{6} \right) \mathcal{W}_0^4 \right) \boxtimes \\
A_{time} = \left(\frac{l_p m_p}{\kappa_1^4} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{W}_0^4 \right) \boxtimes \\
Hz : E_h = \text{watt} \frac{2\pi}{\kappa_1^4} \left(\frac{t_p m_p}{m_e} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{W}_0^4 \right) \boxtimes \\
E_h : Hz = \frac{\kappa_1^4}{2\pi} \left(\frac{m_e}{t_p m_p} \right) \left(1 - \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{W}_0^4 \right) \boxtimes \\
R_{\omega C} = \frac{\kappa_1^4}{4\pi} \left(\frac{m_e}{t_p m_p} \right) \left(1 - \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{W}_0^4 \right) \boxtimes \\
A_{modm} = \kappa_1 \left(\frac{l_p^2 q_p m_p}{t_p m_p} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\mu_B = \frac{\kappa_1}{2} \left(\frac{l_p^2 q_p m_p}{t_p m_e} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\mu_N = \frac{\kappa_1}{2} \left(\frac{l_p^2 q_p m_p}{t_p m_e} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\mu_{tri} = \kappa_1^2 \mathcal{W}_0^2 \left(D_{D0} + \pi \right) \left(\frac{l_p^2 q_p m_p}{t_p m_{tri}} \right) \left(1 - \omega_2 \left(\frac{2\pi}{18} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_{tri}}{\mu_N} = S \left(D_{D0} + \pi \right) \left(\frac{m_e}{m_{tri}} \right) \left(1 + \frac{4}{L_{Li}} \left(\frac{32}{4\pi} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_{tri}}{\mu_N} = \kappa_1 \mathcal{W}_0^2 \left(2D_{D0} + 2\pi \right) \left(\frac{m_e}{m_{tri}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_{tri}}{\mu_B} = \kappa_1 \mathcal{W}_0^2 \left(2D_{D0} + 2\pi \right) \left(\frac{m_e}{m_{tri}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{W}_0^2 \right) \boxtimes \\
g_{tri} = \kappa_1 \mathcal{W}_0^2 \left(4D_{D0} + 4\pi \right) \left(\frac{m_e}{m_{tri}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_m}{\mu_e} = -\frac{S}{C_{CFF}} \left(\frac{m_e}{m_n} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_e}{\mu_n} = -\frac{C_{CFF}}{S} \left(\frac{m_n}{m_e} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{W}_0^2 \right)^{-1} \boxtimes \\
\frac{\mu_n}{\mu_e} = \frac{C_{CFF}}{P_{up}} \left(\frac{m_n}{m_e} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{W}_0^2 \right) \boxtimes \\
\frac{\mu_e}{\mu_e} = \frac{P_{up}}{C_{CFF}} \left(\frac{m_e}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{W}_0^2 \right)^{-1} \boxtimes \\
\frac{\mu_e}{\mu_e} = \left(\frac{m_e}{m_e} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{W}_0^2 \right) \boxtimes \\
G = \left(\frac{l_p^3}{t_p^2 m_p} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{W}_0^3 \right) \boxtimes \\
\bar{G} = \text{GeV}^2 \left(\frac{t_p^4}{l_p^4 m_p^2} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{W}_0^3 \right) \boxtimes \\
m_n = \left(\frac{2\pi^2}{25} \right) \frac{\text{GeV}}{\omega^2} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \left(\frac{2s}{\omega} \right) \mathcal{W}_0^3 \right) \boxtimes \\
m_e = \left(\frac{2\pi^2}{s} \right) \frac{\text{GeV}}{\omega_2} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \frac{\sqrt{2}}{e^{\gamma/3}} \left(\frac{18}{25} \right) \mathcal{W}_0^3 \right) \boxtimes \\
m_p = \frac{2\pi}{35} \left(\frac{4\pi}{6s} \right) (A_{mass}) \left(1 - 18 K_{-6} \mathcal{W}_0^3 \right) \boxtimes \\
\frac{\mu_{de}}{\mu_e} = -P_{up} \left(\frac{2s}{6\pi} \left(\frac{m_e}{m_{de}} \right) \right) \left(1 - 18 \sqrt{L_1} \mathcal{W}_0^3 \right) \boxtimes \\
\frac{\mu_e}{\mu_{de}} = -\frac{1}{P_{up}} \left(\frac{6\pi}{2s} \left(\frac{m_{de}}{m_e}$$

$t_p = 5.39125836832313 \times 10^{-18}$ s	Planck time
$l_p = 1.61629181175645 \times 10^{-35}$ m	Planck length
$\rho_p = 1.87554596713962 \times 10^{-19}$ kg	Planck charge
$T_p = 1.4167869859795 \times 10^{32}$ K	Planck temperature
$m_p = 2.1766268317579 \times 10^{-27}$ kg	Planck mass
$\Xi = 0.199499199973626 \dots \times 10^{-7}$	hyperbolic inversion boundary
$\frac{5}{7}$	number of dimensions break in scale symmetry
$\kappa_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\kappa_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\omega_1 = -1.87646903900417 \dots + 0.06615262615972 \dots i$	3 rd hpc
$\omega_2 = -1.87646903900417 \dots - 0.06615262615972 \dots i$	4 th hpc
$\kappa_0 = 4.08722464916751 \dots$	hyperbolic radius constant
$\kappa_{g0} = 2.00316562310094 \dots$	hyperbolic radian constant
$G_{q0} = 1.01494160640965 \dots$	Gieseking's constant
$V_{\text{te}} = 0.29898321281930 \dots$	figure eight knot hyperbolic volume
$K = 0.9156559417271910 \dots$	Catalan's constant
$c_d = 0.291560904030818 \dots$	domino tiling constant
$D_d = 1.72616281206959 \dots$	dimer constant
$\omega_1 = 0.764977018528595 \dots + 1.32497062714087 \dots i$	omega_1 constant
$\omega_2 = 1.529954037057192 \dots$	omega_2 constant
$i^{(t)}$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\varphi = 1.61803398874989 \dots$	imaginary golden ratio
$\pi = 3.14159265358979 \dots$	golden ratio
$P_{pq} = 2.29557514939263 \dots$	Archimedes' constant
$s = 5.24411510858423 \dots$	universal parabolic constant
$L = 2.62205755429219 \dots$	arc length of the unit lemniscate
$L_1 = 1.31102877174605 \dots$	lemniscate constant
$L_2 = 0.5990707113676796 \dots$	1 st lemniscate constant
$G_0 = 0.847213084793979 \dots$	2 nd lemniscate constant
$C_{q0} = 0.834626841674073 \dots$	ubiquitous constant
$W_e = 0.394373998798109 \dots$	Gauss's constant
$P = 0.998136445958950 \dots$	Weierstrass constant
$M = 1.1100010000000000 \dots$	Ramanujan's 1 st continued fraction constant
$L = -1.74756459463318 \dots$	Liouville's constant
$\mathcal{P} = 1.226742010172035 \dots$	Madelung constant
$e = 2.71828182845904 \dots$	prime constant
$x_{q0} = 2.293166287411536 \dots$	Euler's number
$\gamma = 0.577215664901182 \dots$	1 st Foias constant
$S = 0.828225290871864 \dots$	Euler-Mascheroni constant
$D_{D0} = 0.739985133215157 \dots$	Sierpinski's constant
$L = 0.662743419349181 \dots$	Dotie number
$C_{EPP} = 1.199678640259773 \dots$	Laplace limit
$J_{01} = 2.40482555769572 \dots$	real fixed point of the hyperbolic cotangent
$a_f = 2.50290787509589 \dots$	1 st root of the Bessel function
$\delta_f = 4.66920160910216 \dots$	alpha Feigenbaum constant
$S = 3.24697960371714 \dots$	delta Feigenbaum constant
$P = 1.03127195724474 \dots$	silver constant
$K_{-6} = 0.604543657196732 \dots$	plastic constant
$K_{+6} = 1.15655237442151 \dots$	QRS constant
$T_0 = 2.73150000000000 \dots \times 10^2$ kelvin	Khinchin mean of order −6
$p_0 = 1.00000000000000 \dots \times 10^5$ pascal	freezing point of water
$P_1 = 1.00320000000000 \dots \times 10^5$ pascal	1 bar
$E_1 = 6.019102297911386 \dots \times 10^{14}$ joules	standard atmospheric pressure
$f_1 = 5.44000000000000 \dots \times 10^{14}$ Hz	energy associated with D_{cs}
$\Delta_{\text{peak}} = 4.96511423174427 \dots$	frequency of max/light conversion
$\nu_{\text{peak}} = 2.8214933712207 \dots$	peak 2 of spectral radiance/unit wavelength
	peak emission of spectral flux/unit frequency
$n!$ = factorial function	
$\ln$ = derangement function	$\text{lm} = \text{lumen} = \text{cd} \cdot \text{sr}$
$\log(x)$ = natural logarithm	$N = \text{newton} = \text{kg} \cdot \text{m}/\text{s}^2$
$\Gamma(x)$ = gamma function	$\mathcal{N} = \text{noether} = \text{m} \cdot \text{kg}/\text{s}$
$\zeta(x)$ = Riemann zeta function	$\Omega = \text{ohm} = \text{m}^2 \cdot \text{kg}/\text{s}^2 \cdot \text{C}^2$
	$\text{Pa} = \text{pascal} = \text{kg}/\text{m} \cdot \text{s}^2$
units	$S = \text{siemens} = \text{S}^2/\text{m}^2 \cdot \text{kg} \cdot \text{s}$
m = meter	$T = \text{tesla} = \text{kg}/\text{m} \cdot \text{s}^2 \cdot \text{C}$
s = second	$V = \text{volt} = \text{m}^2 \cdot \text{kg}/\text{s}^2 \cdot \text{C}$
C = coulomb	$W = \text{watt} = \text{m}^2 \cdot \text{kg}/\text{s}^3$
K = kelvin	$W_b = \text{weber} = \text{m}^2 \cdot \text{kg}/\text{s} \cdot \text{C}$
kg = kilogram	
$H = \text{henry} = \text{H}^2 \cdot \text{m}^2 / \text{kg} \cdot \text{C}^2$	
$F = \text{farad} = \text{F}^2 \cdot \text{m}^2 / \text{kg} \cdot \text{C}^2$	
$H = \text{heute} = 1/\text{s}$	
$J = \text{joule} = \text{m}^2 \cdot \text{kg}/\text{s}^2$	
derived units	
$\text{fmHz} = \text{megahertz} = 1.00000000000000 \dots \times 10^{16}$ 1/s	
$\text{fHz} = \text{femtometer} = 1.00000000000000 \dots \times 10^{-15}$ m	
$\text{eV} = \text{electron-volt} = 1.00000000000000 \dots$ eV	
$\text{MeV} = \text{mega-electron-volt} = 1.00000000000000 \dots \times 10^6$ eV	
$\text{GeV} = \text{giga-electron-volt} = 1.00000000000000 \dots \times 10^9$ eV	
$r_n = \left(\frac{\Gamma(n)}{4}\right)^{-2}$	$r_n = K$
$r_n = K$	$r_n = \sqrt{\frac{C_4}{U}}$
$r_n = K$	